Chapter 10. Analysis of Variance and Regression

One-way and two-way ANOVA, quadratic forms, linear regression

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10.1 One-way analysis of variance

The **one-way normal model** (Chapter 4) is

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad i = 1, \dots, k,\ j = 1, \dots, n_i,$$

for the effect of a single **factor** with k **levels**; applying level i is called **treatment** i. With $n = \sum n_i$, $\bar{\mu} = \sum_i n_i \mu_i / n$ (overall effect) and $\alpha_i = \mu_i - \bar{\mu}$ (effect of treatment i), it can be written

$$X_{ij} = \bar{\mu} + \alpha_i + e_{ij}, \qquad \sum_i n_i \alpha_i = 0.$$

Theorem 10.1 (UMVUEs)

In the one-way normal model, the UMVUEs are

$$\hat{\mu}_i = \bar{X}_i = \frac{1}{n_i} \sum_j X_{ij}, \qquad \hat{\bar{\mu}} = \sum_i \frac{n_i}{n} \hat{\mu}_i, \qquad \hat{\alpha}_i = \hat{\mu}_i - \hat{\bar{\mu}}, \qquad \hat{\sigma}^2 = \frac{1}{n-k} \sum_{i,j} (X_{ij} - \bar{X}_i)^2.$$

Proof. The joint pdf is $\exp\{-\frac{1}{2\sigma^2}\sum_{i,j}(x_{ij}-\mu_i)^2 - \frac{n}{2}\log(2\pi\sigma^2)\}$, a $(k+1)$-parameter exponential family satisfying the conditions of Theorem 8.8. So $(\bar{X}_1, \dots, \bar{X}_k, \sum_{i,j}(X_{ij} - \bar{X}_i)^2)$ is complete sufficient. The listed estimators are unbiased functions of it (Theorem 4.9 in Volume I for $\hat{\sigma}^2$), hence UMVUEs by Lehmann–Scheffé. ■

The MLE of σ^2 is $\sum (X_{ij} - \bar{X}_i)^2 / n$; the other MLEs coincide with the UMVUEs.

Theorem 10.2 (Sampling distributions)

(a) $\hat{\mu}_i = \bar{X}_i$ are independent, $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$.
(b) $\hat{\sigma}^2$ is independent of $(\bar{X}_1, \dots, \bar{X}_k)$.
(c) $(n-k)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-k)$.

Proof. Apply Theorem 4.5 in Volume I to each group: $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$ is independent of $SS_i = \sum_j (X_{ij} - \bar{X}_i)^2$, and $SS_i/\sigma^2 \sim \chi^2(n_i - 1)$. The groups are independent and $(n-k)\hat{\sigma}^2 = \sum_i SS_i$; add the chi-squares. ■

Simultaneous confidence intervals

Write $\hat{\mu} = (\hat{\mu}_1, \dots, \hat{\mu}_k)'$, $D = \mathrm{Var}(\hat{\mu})/\sigma^2 = \mathrm{diag}(1/n_i)$, and for a $k \times r$ matrix C of rank r let $\mathrm{col}(C) = \{C\mathbf{a} : \mathbf{a} \in \mathbb{R}^r\}$.

Theorem 10.3 (Confidence sets and Scheffé intervals)

(a) $P\{(C'\mu - C'\hat{\mu})'(C'DC)^{-1}(C'\mu - C'\hat{\mu}) \le r\hat{\sigma}^2 F_\alpha(r, n-k)\} = 1 - \alpha$.

(b) $P\Big\{|\mathbf{c}'\mu - \mathbf{c}'\hat{\mu}| \le \sqrt{\textstyle\sum_i c_i^2 \hat{\sigma}^2/n_i}\sqrt{r F_\alpha(r, n-k)} \ \text{ for all } \mathbf{c} \in \mathrm{col}(C)\Big\} = 1 - \alpha$.

Proof. (a) $C'\hat{\mu} \sim N(C'\mu, \sigma^2 C'DC)$, so $(C'\hat{\mu} - C'\mu)'(C'DC)^{-1}(C'\hat{\mu} - C'\mu)/\sigma^2 \sim \chi^2(r)$ (Theorem 4.18 in Volume I). It is independent of $(n-k)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-k)$, and the ratio of the two, each divided by its degrees of freedom, is $F(r, n-k)$.

(b) For $\mathbf{c} = C\mathbf{a}$, with $\mathbf{d} = C'(\mu - \hat{\mu})$ and $M = C'DC$, the Cauchy–Schwarz inequality gives

$$\max_{\mathbf{c} \in \mathrm{col}(C), \mathbf{c} \neq \mathbf{0}} \frac{\{\mathbf{c}'(\mu - \hat{\mu})\}^2}{\mathbf{c}' D \mathbf{c}} = \max_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}'\mathbf{d})^2}{\mathbf{a}' M \mathbf{a}} = \max_{\mathbf{b} \neq \mathbf{0}} \frac{(\mathbf{b}' M^{-1/2} \mathbf{d})^2}{\mathbf{b}'\mathbf{b}} = \mathbf{d}' M^{-1} \mathbf{d}.$$

So the event in (b) is the same as the event in (a). ■

A linear combination $\sum c_i\mu_i$ with $\sum c_i = 0$ is a **contrast**; the simplest are the pairwise differences $\mu_i - \mu_l = \alpha_i - \alpha_l$. The contrasts form a space of dimension $k-1$, so:

Theorem 10.4 (Simultaneous intervals for contrasts)

(a) (Scheffé) With probability $1-\alpha$, for all $\mathbf{c}$ with $\sum c_i = 0$,

$$|\mathbf{c}'\alpha - \mathbf{c}'\hat{\alpha}| \le \sqrt{\textstyle\sum_i c_i^2\hat{\sigma}^2/n_i}\,\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(b) In particular, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i-\alpha_l)-(\hat{\alpha}_i-\hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(c) (Bonferroni) With $m = k(k-1)/2$ and $\alpha^* = \alpha/m$, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i-\alpha_l)-(\hat{\alpha}_i-\hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\; t_{\alpha^*/2}(n-k).$$

Proof. (a) is Theorem 10.3(b) with $\mathrm{col}(C)$ the $(k-1)$-dimensional space of contrasts;

(b) is a sub-event.

(c) For a single pair, $r=1$ in Theorem 10.3, and $F_{\alpha^*}(1,n-k) = t_{\alpha^*/2}(n-k)^2$, so each interval has coverage $1-\alpha^*$. By the Bonferroni inequality $P(A_1\cap\cdots\cap A_m) \ge 1-\sum_i(1-P(A_i)) = 1-m\alpha^* = 1-\alpha$. ■

When there are few comparisons, Bonferroni intervals are shorter; Scheffé intervals cover all contrasts at once. For example, with $n=20$ and $\alpha=0.05$: for $k=3$ the multipliers are $\sqrt{(k-1)F_\alpha} = 2.680$ and $t_{\alpha^*/2} = 2.654$; for $k=7$, 4.182 versus 3.760.

Testing equality of treatments

Theorem 10.5 (F test)

Let $SSB = \sum_i n_i(\bar{X}_i - \bar{X})^2$ (between groups) and $SSW = \sum_{i,j}(X_{ij}-\bar{X}_i)^2 = (n-k)\hat{\sigma}^2$ (within groups). The LRT of $H_0: \mu_1 = \cdots = \mu_k$ (i.e. $\alpha_1 = \cdots = \alpha_k = 0$) rejects when

$$F_n = \frac{SSB/(k-1)}{SSW/(n-k)} \ge F_\alpha(k-1,n-k).$$

Proof. The MLEs are $\hat{\theta}^\Omega = (\bar{x}_1,\dots,\bar{x}_k,\hat{\sigma}^{2,\Omega})$ with $n\hat{\sigma}^{2,\Omega} = SSW$, and $\hat{\theta}^{\Omega_0} = (\bar{x},\dots,\bar{x},\hat{\sigma}^{2,0})$ with $n\hat{\sigma}^{2,0} = \sum_{i,j}(x_{ij}-\bar{x})^2 = SSW + SSB$. So

$$2\{l(\hat{\theta}^\Omega) - l(\hat{\theta}^{\Omega_0})\} = n\log\frac{\hat{\sigma}^{2,0}}{\hat{\sigma}^{2,\Omega}} = n\log\Big(1+\frac{SSB}{SSW}\Big),$$

increasing in F_n. For the null distribution, write $\mathbf{X} = (X_{ij})$ as an n-vector and let A map it to the vector with entries $\bar{X}_i - \bar{X}$. Then A is symmetric and idempotent, $SSB = \mathbf{X}'A\mathbf{X}$, and $\mathrm{tr}A = \sum_{i,j}(1/n_i - 1/n) = k-1$. Under H_0, $\mathbf{Z} = (\mathbf{X}-\bar{\mu}\mathbf{1})/\sigma \sim N_n(\mathbf{0},I)$ and $A\mathbf{1} = \mathbf{0}$, so $SSB/\sigma^2 = \mathbf{Z}'A\mathbf{Z} \sim \chi^2(k-1)$. Since SSB is a function of the group means, it is independent of $SSW/\sigma^2 \sim \chi^2(n-k)$ (Theorem 10.2). Hence $F_n \sim F(k-1,n-k)$ under H_0. ■

One sometimes uses the unweighted effects $\tilde{\alpha}_i = \mu_i - \tilde{\mu}$, $\tilde{\mu} = \sum\mu_i/k$. Since $\tilde{\alpha}_1 = \cdots = \tilde{\alpha}_k = 0$ is again $\mu_1 = \cdots = \mu_k$, the same F test applies, and for contrasts $\sum c_i\tilde{\alpha}_i = \sum c_i\mu_i = \sum c_i\alpha_i$.

Example 10.1

Yields (in kg) of a crop under three fertilizers:

fertilizer	observations	n_i	$\bar{x}_i$
A	20, 22, 19, 23	4	21
B	25, 27, 26	3	26
C	22, 21, 24, 23, 25	5	23

Here $n = 12$, $\bar{x} = (84 + 78 + 115)/12 = 23.083$, $SSB = 4(2.083)^2 + 3(2.917)^2 + 5(0.083)^2 = 42.92$, and $SSW = 10 + 2 + 10 = 22$. So $F = (42.92/2)/(22/9) = 8.78 > F_{0.05}(2, 9) = 4.26$: the fertilizers differ. Bonferroni intervals ($m = 3$, $t_{0.05/6}(9) = 2.933$, $\hat{\sigma}^2 = 2.444$) for $\mu_B - \mu_A$: $5 \pm 2.933\sqrt{2.444(1/3 + 1/4)} = 5 \pm 3.50$, which excludes 0.

10.2 Two-way analysis of variance

With two factors A (levels $i = 1, \dots, a$) and B (levels $j = 1, \dots, b$), each combination (i, j) is a **cell**, observed n_{ij} times:

$$X_{ijk} = \mu_{ij} + e_{ijk}, \qquad e_{ijk} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad k = 1, \dots, n_{ij}.$$

This is a one-way model with ab treatments, so as in Section 10.1:

Theorem 10.6

(a) The UMVUEs are $\hat{\mu}_{ij} = \bar{X}_{ij}$ and $\hat{\sigma}^2 = \sum_{i,j,k}(X_{ijk} - \bar{X}_{ij})^2/(n - ab)$, $n = \sum n_{ij}$.
(b) The $\bar{X}_{ij}$ are independent $N(\mu_{ij}, \sigma^2/n_{ij})$.
(c) $\hat{\sigma}^2$ is independent of the cell means.
(d) $(n - ab)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n - ab)$.
(e) Confidence sets and Scheffé intervals are as in Theorem 10.3, with $D = \operatorname{diag}(1/n_{ij})$ and $n - ab$ degrees of freedom.

The effects are defined by

$$\bar{\mu}_{\cdot\cdot} = \sum_{i,j} \frac{n_{ij}}{n}\mu_{ij}, \quad \bar{\mu}_{i\cdot} = \sum_{j} \frac{n_{ij}}{n_{i\cdot}}\mu_{ij}, \quad \bar{\mu}_{\cdot j} = \sum_{i} \frac{n_{ij}}{n_{\cdot j}}\mu_{ij},$$

$$\alpha_i = \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}, \qquad \beta_j = \bar{\mu}_{\cdot j} - \bar{\mu}_{\cdot\cdot}, \qquad \gamma_{ij} = \mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}.$$

α_i and β_j are the **main effects** of A and B; γ_{ij}, the part of μ_{ij} not explained additively, is the **interaction**. The model is **balanced** if $n_{ij} \equiv r$. Then the model reads

$$X_{ijk} = \bar{\mu}_{\cdot\cdot} + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad \sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0.$$

Theorem 10.7 (Pairwise comparisons of main effects)

In the balanced model, $\hat{\alpha}_i = \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots}$, $n_{i\cdot} = rb$, $n = rab$, and

(a) with probability at least $1 - \alpha$, for all $i \neq l$, $|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \leq \sqrt{2\hat{\sigma}^2/(rb)}\sqrt{(a-1)F_\alpha(a-1, n-ab)}$;
(b) with $m = a(a-1)/2$, $\alpha^* = \alpha/m$, the same holds with $t_{\alpha^*/2}(n - ab)$ in place of $\sqrt{(a-1)F_\alpha(a-1, n-ab)}$.

Proof. $\alpha_i - \alpha_l = \sum_j(\mu_{ij} - \mu_{lj})/b$ is a contrast in the μ_{ij}, with $\sum c^2/n_{ij} = 2/(rb)$; the space of such contrasts has dimension $a-1$. Apply Theorem 10.6(e) and the Bonferroni argument of Theorem 10.4. ■

Theorem 10.8 (Test for interaction)

In the balanced model let

$$SS_{AB} = \sum_{i,j} r(\bar{X}_{ij\cdot} - \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdot j\cdot} + \bar{X}_{\cdots})^2, \qquad SSE = \sum_{i,j,k}(X_{ijk} - \bar{X}_{ij\cdot})^2.$$

The LRT of $H_0^{AB}: \gamma_{ij} = 0$ for all i, j (the additive model) rejects when

$$F_n = \frac{SS_{AB}/\{(a-1)(b-1)\}}{SSE/(n-ab)} \geq F_\alpha((a-1)(b-1), n-ab).$$

Proof. Under Ω, $\hat{\mu}_{ij} = \bar{x}_{ij\cdot}$ and $n\hat{\sigma}^{2,\Omega} = SSE$. Under H_0^{AB}, $\mu_{ij} = \bar{\mu} + \alpha_i + \beta_j$, and in the balanced case the least squares fit is $\hat{\mu}^0_{ij} = \bar{x}_{i\cdot\cdot} + \bar{x}_{\cdot j\cdot} - \bar{x}_{\cdots}$. By orthogonality of the decomposition, $n\hat{\sigma}^{2,0} = SSE + SS_{AB}$, so $2\{l(\hat{\theta}^{\Omega}) - l(\hat{\theta}^{\Omega_0})\} = n\log(1 + SS_{AB}/SSE)$, increasing in F_n. For the null distribution, the map $\Pi : (x_{ijk}) \mapsto (\bar{x}_{ij\cdot} - \bar{x}_{i\cdot\cdot} - \bar{x}_{\cdot j\cdot} + \bar{x}_{\cdots})$ is symmetric and idempotent, $SS_{AB} = \mathbf{X}'\Pi\mathbf{X}$, and

$$\mathrm{tr}\Pi = \sum_{i,j,k}\left(\frac{1}{r} - \frac{1}{rb} - \frac{1}{ra} + \frac{1}{rab}\right) = ab - a - b + 1 = (a-1)(b-1).$$

Under H_0^{AB}, Π annihilates the mean vector, so $SS_{AB}/\sigma^2 \sim \chi^2((a-1)(b-1))$. It depends only on the cell means, so it is independent of $SSE/\sigma^2 \sim \chi^2(n-ab)$. ■

Theorem 10.9 (Tests for main effects)

In the balanced model, with $SS_A = \sum_i br(\bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots})^2$ and $SS_B = \sum_j ar(\bar{X}_{\cdot j\cdot} - \bar{X}_{\cdots})^2$:

(a) the LRT of $H_0^A: \alpha_1 = \cdots = \alpha_a = 0$ rejects when $\dfrac{SS_A/(a-1)}{SSE/(n-ab)} \geq F_\alpha(a-1, n-ab)$;

(b) the LRT of $H_0^B: \beta_1 = \cdots = \beta_b = 0$ rejects when $\dfrac{SS_B/(b-1)}{SSE/(n-ab)} \geq F_\alpha(b-1, n-ab)$.

The proofs follow that of Theorem 10.8. The results are summarized in the two-way ANOVA table; δ is the noncentrality parameter of the next section.

source	SS	df	MS	F	δ
A	SS_A	$a-1$	MS_A	MS_A/MSE	$\sum br\alpha_i^2/\sigma^2$
B	SS_B	$b-1$	MS_B	MS_B/MSE	$\sum ar\beta_j^2/\sigma^2$
$A \times B$	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	MS_{AB}/MSE	$\sum r\gamma_{ij}^2/\sigma^2$
error	SSE	$n-ab$	MSE		
total	SST	$n-1$			

10.3 Power of F tests

Motivation. A test is useful only if it can detect effects of a relevant size. Under the alternative the F statistic has a noncentral F distribution, whose noncentrality parameter determines the power and the required sample size.

Under alternatives the F statistics have **noncentral** distributions.

Theorem 10.10 (Noncentral chi-square)

Let $X_i \sim N(\mu_i, 1)$, $i = 1, \dots, r$, be independent, $Y = X_1^2 + \cdots + X_r^2$ and $\delta = \mu_1^2 + \cdots + \mu_r^2$. Then

(a) $K_Y(t) = -\frac{r}{2}\log(1-2t) + \frac{\delta}{2}\cdot\frac{2t}{1-2t} = \sum_{k\ge 1}\frac{t^k}{k!}2^{k-1}(k-1)!\,(r + k\delta)$ for $t < 1/2$;

(b) with $p_{P(\delta/2)}(k)$ the Poisson$(\delta/2)$ pmf, $f_Y(y) = \sum_{k\ge 0} p_{P(\delta/2)}(k)\, f_{\chi^2(r+2k)}(y)$.

In particular the law of Y depends on $(\mu_1, \dots, \mu_r)$ only through δ. It is the **noncentral chi-square** $\chi^2(r; \delta)$.

Proof. (a) For $X \sim N(\mu, 1)$ and $t < 1/2$, completing the square,

$$Ee^{tX^2} = \int \frac{1}{\sqrt{2\pi}} \exp\Big\{-\frac{1}{2}(1-2t)\Big(x - \frac{\mu}{1-2t}\Big)^2 + \frac{t\mu^2}{1-2t}\Big\}dx = (1-2t)^{-1/2}\exp\Big(\frac{\mu^2 t}{1-2t}\Big).$$

Multiply over i and take logs. Expanding $-\log(1-2t) = \sum (2t)^k/k$ and $\frac{2t}{1-2t} = \sum_{k\ge 1}(2t)^k$ gives the series.

(b) By (a), $Y \overset{d}{=} V + Z^2$ with $V \sim \chi^2(r-1)$ independent of $Z \sim N(\sqrt{\delta}, 1)$. With $\theta = \sqrt{\delta}$,

$$f_{Z^2}(x) = \frac{1}{2\sqrt{x}}\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x - \frac{1}{2}\theta^2}\big(e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}}\big) = \sum_{k\ge 0} p_{P(\delta/2)}(k) f_{\chi^2(1+2k)}(x),$$

by expanding $e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}} = 2\sum_k \theta^{2k}x^k/(2k)!$ and using $\Gamma(\frac{1}{2})\Gamma(2k+1) = 2^{2k}\Gamma(k+\frac{1}{2})\Gamma(k+1)$. Convolving with $f_{\chi^2(r-1)}$ and using $\chi^2(r-1) \oplus \chi^2(1+2k) = \chi^2(r+2k)$ gives (b). ∎

Theorem 10.11

If $\mathbf{X} \sim N(\mu, I)$ and A is symmetric idempotent, then $\mathbf{X}'A\mathbf{X} \sim \chi^2(r; \delta)$ with $r = \mathrm{tr}A$ and $\delta = \mu' A \mu$.

Proof. Write $A = P\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P'$ with P orthogonal, and $\mathbf{Z} = P'\mathbf{X} \sim N(\eta, I)$, $\eta = P'\mu$. Then $\mathbf{X}'A\mathbf{X} = Z_1^2 + \cdots + Z_r^2 \sim \chi^2(r; \eta_1^2 + \cdots + \eta_r^2)$, and $\eta_1^2 + \cdots + \eta_r^2 = \mu' A \mu$. ∎

If $V_1 \sim \chi^2(r_1; \delta)$ and $V_2 \sim \chi^2(r_2)$ are independent, $(V_1/r_1)/(V_2/r_2)$ has the **noncentral** F distribution $F(r_1, r_2; \delta)$.

Example 10.2 (Power of the one-way F test)

In the one-way model, $SSB = \mathbf{X}'A\mathbf{X}$ with A idempotent of trace $k-1$, and A maps the mean vector (μ_i) to (α_i). By Theorem 10.11,

$$SSB/\sigma^2 \sim \chi^2(k-1; \delta), \qquad \delta = \sum_i n_i \alpha_i^2/\sigma^2,$$

so $F_n \sim F(k-1, n-k; \delta)$, and the power is $P\{F(k-1, n-k; \delta) \ge F_\alpha(k-1, n-k)\}$. It increases in δ: writing $F_n \overset{d}{=} \frac{(V+Z^2)/(k-1)}{W/(n-k)}$ with $V \sim \chi^2(k-2)$, $Z \sim N(\sqrt{\delta}, 1)$, $W \sim \chi^2(n-k)$ independent, the power is $P_\theta(Z^2 \ge Y)$ with Y independent of Z and $\theta = \sqrt{\delta}$, and

$$\frac{d}{d\theta} P_\theta(Z^2 \ge y) = e^{-y/2}\phi(\theta)\big(e^{\sqrt{y}\theta} - e^{-\sqrt{y}\theta}\big) > 0 \qquad (\theta > 0).$$

In the same way, in the balanced two-way model the interaction test has power $P\{F((a-1)(b-1), n-ab; \delta_{AB}) \geq F_\alpha\}$ with $\delta_{AB} = \sum r\gamma_{ij}^2/\sigma^2$, increasing in δ_{AB}.

10.4 Regression analysis

Motivation. Regression is the same linear-model theory as ANOVA with continuous regressors. Least squares estimates, their normal-theory distributions and the F-test for the slope follow from the results of the previous sections.

The normal linear regression model (Chapters 4 and 6) is

$$\mathbf{Y} = X\beta + \mathbf{e}, \qquad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \qquad \beta \in \mathbb{R}^{p+1},\ \sigma^2 > 0,\ \operatorname{rank} X = p+1.$$

Theorem 10.12

(a) The UMVUEs of $\mathbf{c}'\beta$ and σ^2 are $\mathbf{c}'\hat\beta = \mathbf{c}'(X'X)^{-1}X'\mathbf{Y}$ and $\hat\sigma^2 = \|\mathbf{Y} - X\hat\beta\|^2/(n-p-1)$.

(b) $\hat\beta \sim N_{p+1}(\beta, \sigma^2(X'X)^{-1})$.

(c) $\hat\beta$ and $\hat\sigma^2$ are independent.

(d) $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$.

Proof. The density $\exp\{-\mathbf{y}'\mathbf{y}/(2\sigma^2) + \mathbf{y}'X\beta/\sigma^2 - \beta'X'X\beta/(2\sigma^2) - \frac{n}{2}\log(2\pi\sigma^2)\}$ is a full-rank exponential family, so $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ is complete sufficient. Since $X'\mathbf{Y} = (X'X)\hat\beta$ and $\mathbf{Y}'\mathbf{Y} = (n-p-1)\hat\sigma^2 + \hat\beta'(X'X)\hat\beta$, $(\hat\beta, \hat\sigma^2)$ is a one-to-one function of it, and the unbiased estimators $\mathbf{c}'\hat\beta$, $\hat\sigma^2$ (Theorem 6.16) are UMVUEs. (b)–(d) are Theorem 4.19 in Volume I. ∎

Theorem 10.13 (Confidence sets for β)

For a $(p+1)\times r$ matrix C of rank r:

(a) $P\{(C'\beta - C'\hat\beta)'(C'(X'X)^{-1}C)^{-1}(C'\beta - C'\hat\beta) \leq r\hat\sigma^2 F_\alpha(r, n-p-1)\} = 1-\alpha$;

(b) $P\Big\{|\mathbf{c}'\beta - \mathbf{c}'\hat\beta| \leq \sqrt{\mathbf{c}'(X'X)^{-1}\mathbf{c}\,\hat\sigma^2}\sqrt{rF_\alpha(r, n-p-1)}\ \ \forall \mathbf{c} \in \operatorname{col}(C)\Big\} = 1-\alpha$.

The proof is that of Theorem 10.3.

Often only some coefficients matter; e.g. with an intercept, whether the slopes $\beta_1, \dots, \beta_p$ are zero. Partition

$$\mathbf{Y} = X_0\beta_0 + X_1\beta_1 + \mathbf{e}, \qquad \operatorname{rank} X_0 = p_0,\ \operatorname{rank}(X_0, X_1) = p_0 + p_1,$$

and let $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$, $X_{1|0} = (I - \Pi_0)X_1$, $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$ and $\Pi_{0,1} = X(X'X)^{-1}X'$. From Theorem 6.17, $\Pi_{0,1} = \Pi_0 + \Pi_{1|0}$ and $\Pi_0\Pi_{1|0} = 0$.

Theorem 10.14 (Testing a subset of coefficients)

For $H_0 : \beta_1 = \mathbf{0}$ vs $H_1 : \beta_1 \neq \mathbf{0}$:

(a) $\min_{\beta_0}\|\mathbf{Y} - X_0\beta_0\|^2 = \min_{\beta_0,\beta_1}\|\mathbf{Y} - X_0\beta_0 - X_1\beta_1\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$.

(b) With $R(1|0) = \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$ and $SSE = \mathbf{Y}'(I - \Pi_0 - \Pi_{1|0})\mathbf{Y}$, the LRT rejects when

$$F_n = \frac{R(1|0)/p_1}{SSE/(n-p_0-p_1)} \geq F_\alpha(p_1, n-p_0-p_1).$$

(c) $F_n \sim F(p_1, n-p_0-p_1; \delta)$ with $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$, and the power increases in δ.

Proof. (a) The minima are $\mathbf{Y}'(I-\Pi_0)\mathbf{Y}$ and $\mathbf{Y}'(I-\Pi_{0,1})\mathbf{Y}$, and $I-\Pi_0=(I-\Pi_{0,1})+\Pi_{1|0}$.

(b) The MLEs of σ^2 are SSE/n under Ω and $(SSE+R(1|0))/n$ under H_0, so $2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}=n\log(1+R(1|0)/SSE)$, increasing in F_n.

(c) $\Pi_{1|0}$ is idempotent with $\mathrm{tr}\Pi_{1|0}=\mathrm{tr}\{(X_{1|0}'X_{1|0})^{-1}X_{1|0}'X_{1|0}\}=p_1$, so $R(1|0)/\sigma^2\sim\chi^2(p_1;\delta)$ with $\delta=(X_0\beta_0+X_1\beta_1)'\Pi_{1|0}(X_0\beta_0+X_1\beta_1)/\sigma^2=\beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$ since $\Pi_{1|0}X_0=0$. Also $SSE/\sigma^2\sim\chi^2(n-p_0-p_1)$, and $\Pi_{1|0}(I-\Pi_0-\Pi_{1|0})=0$ gives independence. Under H_0, $\delta=0$. Monotonicity of the power is shown as in Example 10.2. ■

Theorem 10.15 (General linear hypotheses)

For a $(p+1)\times r$ matrix C of rank r, the LRT of $H_0: C'\beta=\mathbf{0}$ vs $H_1: C'\beta\neq\mathbf{0}$ satisfies:

(a) with $\Pi_{1|0}=X(X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'(X'X)^{-1}X'$,

$$\min_{C'\beta=\mathbf{0}}\|\mathbf{Y}-X\beta\|^2=\min_{\beta}\|\mathbf{Y}-X\beta\|^2+\mathbf{Y}'\Pi_{1|0}\mathbf{Y},$$

$$\mathbf{Y}'\Pi_{1|0}\mathbf{Y}=(C'\hat\beta)'\{C'(X'X)^{-1}C\}^{-1}(C'\hat\beta);$$

(b) with $SSE=\mathbf{Y}'(I-X(X'X)^{-1}X')\mathbf{Y}$, it rejects when $\dfrac{\mathbf{Y}'\Pi_{1|0}\mathbf{Y}/r}{SSE/(n-p-1)}\geq F_\alpha(r,n-p-1)$;

(c) this statistic is $F(r,n-p-1;\delta)$ with $\delta=\beta'C\{C'(X'X)^{-1}C\}^{-1}C'\beta/\sigma^2$.

Proof. Reduce to Theorem 10.14. Choose C_1 ($(p+1)\times r$) with orthonormal columns spanning $\mathrm{col}(C)$, and extend to an orthonormal basis $(C_0\ C_1)$ of $\mathbb{R}^{p+1}$; then $C=C_1B$ for an invertible B. With $D_0=XC_0$, $D_1=XC_1$, $\gamma_0=C_0'\beta$, $\gamma_1=C_1'\beta$,

$$X\beta=X(C_0\ C_1)(C_0\ C_1)'\beta=D_0\gamma_0+D_1\gamma_1,$$

and $C'\beta=\mathbf{0}\iff\gamma_1=\mathbf{0}$. Theorem 10.14 applies, and $\Pi_{1|0}$ is the projection onto the orthogonal complement of $\mathrm{col}(D_0)$ in $\mathrm{col}(X)$:

$$\{X\beta:\beta'X'XC_0=\mathbf{0}\}=\{X\beta:X'X\beta\in\mathrm{col}(C)\}=\mathrm{col}(X(X'X)^{-1}C).$$

With $S=X(X'X)^{-1}C$, $\Pi_{1|0}=S(S'S)^{-1}S'$, which is the stated matrix, and $S'\mathbf{Y}=C'\hat\beta$ gives the quadratic form. ■

Linear models. The ANOVA models are regression models whose design matrix has dependent columns, made identifiable by linear constraints:

$$\mathbf{Y}=X\beta+\mathbf{e},\quad C'\beta=\mathbf{0},\quad \mathbf{e}\sim N_n(\mathbf{0},\sigma^2I),\quad \mathrm{rank}\,X=p+1-r<p+1,$$

where the rows of X and C' together are linearly independent. Such a model is a **linear model**.

Example 10.3 (One-way model as a linear model)

In $X_{ij}=\bar\mu+\alpha_i+e_{ij}$, $\sum n_i\alpha_i=0$, stack $\mathbf{Y}=(X_{11},\dots,X_{1n_1},\dots,X_{k1},\dots,X_{kn_k})'$, $\beta=(\bar\mu,\alpha_1,\dots,\alpha_k)'$, $\mathbf{c}=(0,n_1,\dots,n_k)'$, and

$$X=\begin{pmatrix}\mathbf{1}_{n_1}&\mathbf{1}_{n_1}&\mathbf{0}&\cdots&\mathbf{0}\\ \mathbf{1}_{n_2}&\mathbf{0}&\mathbf{1}_{n_2}&\cdots&\mathbf{0}\\ \vdots&&&\ddots&\\ \mathbf{1}_{n_k}&\mathbf{0}&\mathbf{0}&\cdots&\mathbf{1}_{n_k}\end{pmatrix}.$$

This **design matrix** has rank $k < k+1$; the constraint $\mathbf{c}'\beta = 0$ makes β identifiable. Since $X'X$ is singular, formulas must be written with projections rather than $(X'X)^{-1}$, but the estimation and testing theory goes through as in Theorem 10.15.